Name of the Student: G. Manasa	**Academic Year:** 2024-28
Student Registration Number: 241U1R3007	**Year & Term:** II&I
Study Level: UG	**Class & Section:** CSE-DS
Name of the Course: Digital Electronics	**Name of the Instructor:** Nuzhath Farhana
Name of the Assessment: Lab Report-3	**Date of Submission:** 03-08-2025

Lab Report 3

Simulation of Boolean Expressions using Logic Gates

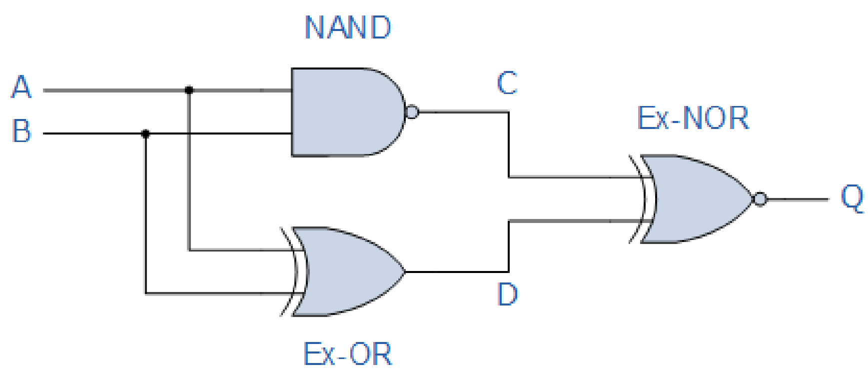
Objective: To implement and verify the functionality of Boolean expressions using basic logic gates in SimulIDE.

Software Used: SimulIDE

Circuit Diagrams:

Boolean Algebra Examples No1

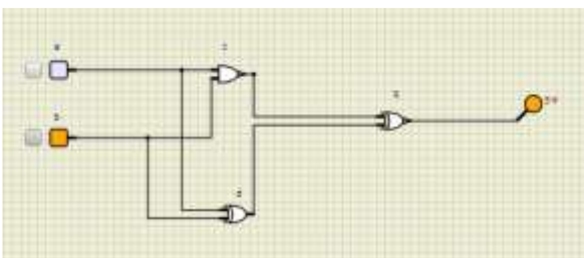
Construct a Truth Table for the logical functions at points C, D and Q in the following circuit and

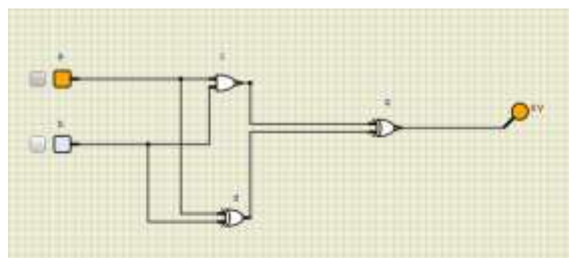


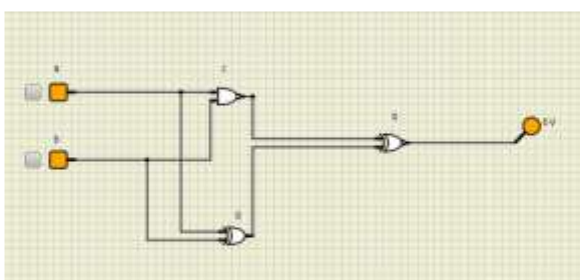
identify a single logic gate that can be used to replace the whole circuit.

Inputs		Output at		
A	B	C	D	Q
0	0	1	0	0
0	1	1	1	1
1	0	1	1	1
1	1	0	0	1

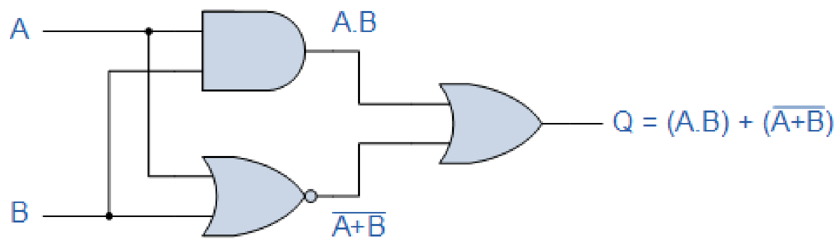
Circuit Executions







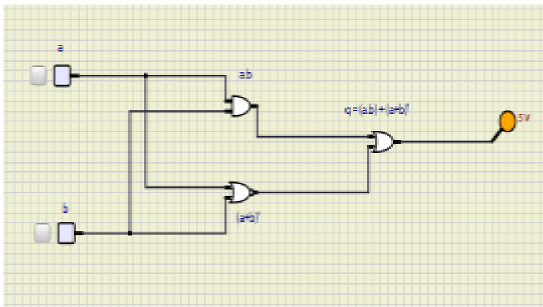
Boolean Algebra Examples

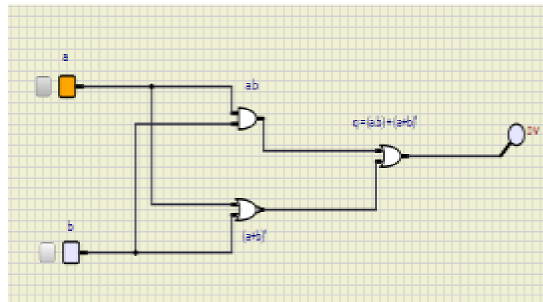


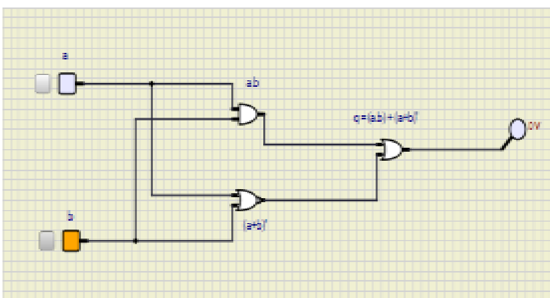
The output of the system is given as Q = (A.B) + (A+B)' but the notation A'+B' is the same as the De Morgan´s notation A'.B', Then substituting A'.B' into the output expression gives us a final output notation of Q = (A.B)+(A'.B'), which is the Boolean notation for an Exclusive- NOR Gate

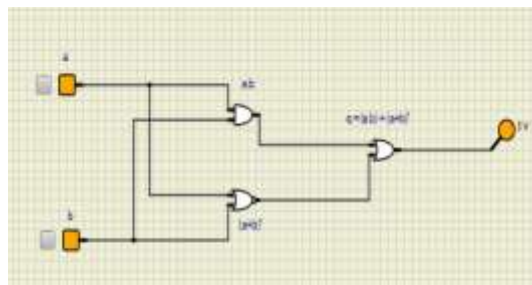
Inputs		Intermediates		Output
B	A	A.B	$\overline{A+B}$	Q
0	0	0	1	1
0	1	0	0	0
1	0	0	0	0
1	1	1	0	1

Circuit Executions:







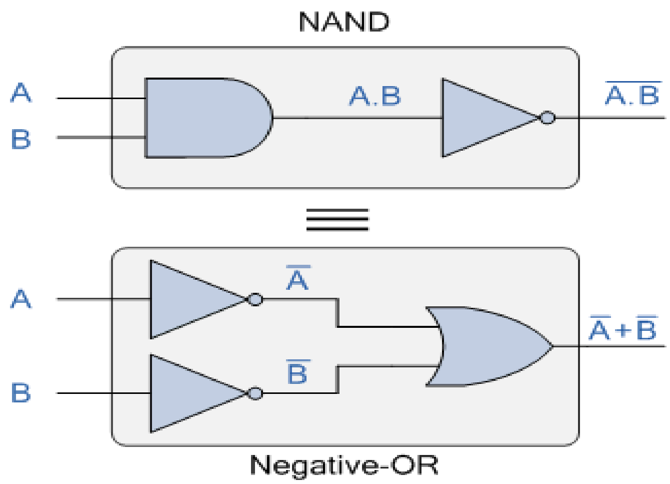


DeMorgan's Theorem

DeMorgan´s Theorem and Laws can be used to to find the equivalency of the NAND and NOR gates

DeMorgan's First Theorem

DeMorgan's First theorem proves that when two (or more) input variables are AND'ed and negated, they are equivalent to the OR of the complements of the individual variables. Thus the equivalent of the NAND function will be a negative-OR function, proving that A.B = A+B. We can show this operation using the following table.

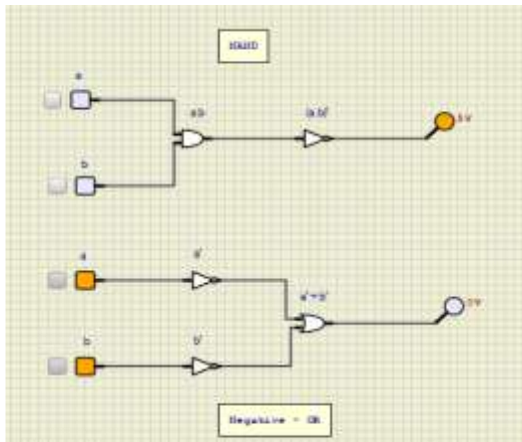


Verifying DeMorgan's First Theorem using Truth Table

Inputs		Truth Table Outputs for Each Term				
B	A	A. B	A. B	A	B	A + B
0	0	0	**1**	1	1	**1**
0	1	0	**1**	0	1	**1**
1	0	0	**1**	1	0	**1**
1	1	1	**0**	0	0	**0**

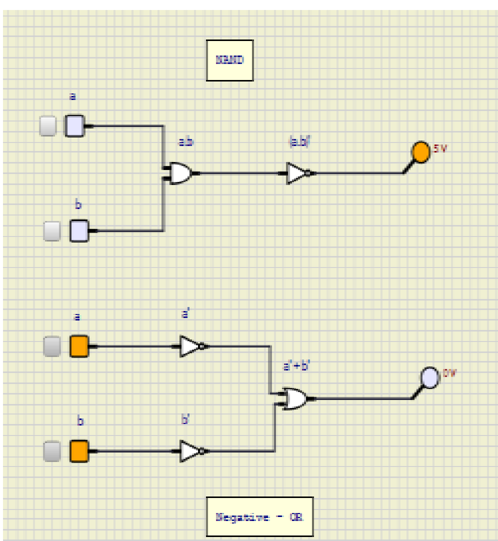
We can also show that A.B = A+B using logic gates as shown.

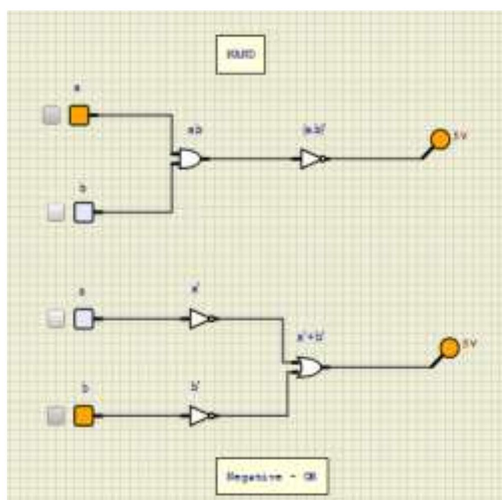
Circuit Executions:



We can also show that A.B = A+B using logic gates as shown.

Circuit Executions:

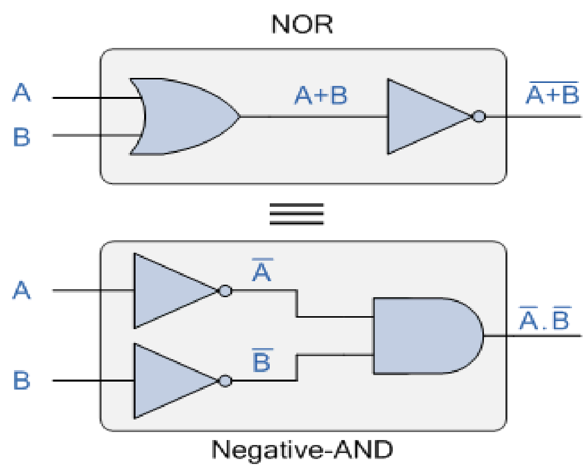




DeMorgan's Second Theorem

DeMorgan's Second theorem proves that when two (or more) input variables are OR'ed and negated, they are equivalent to the AND of the complements of the individual variables. Thus the equivalent of the NOR function is a negative-AND function proving that A+B = A.B, and again we can show operation this using the following truth table.

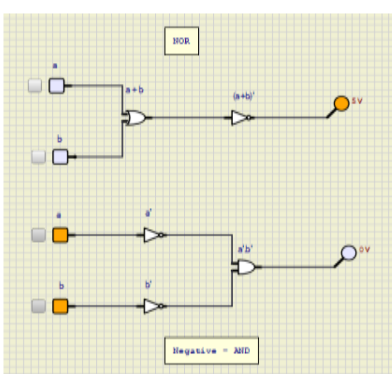
DeMorgan's Second Law Implementation using Logic Gates

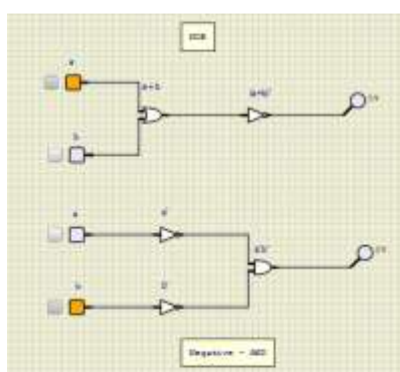


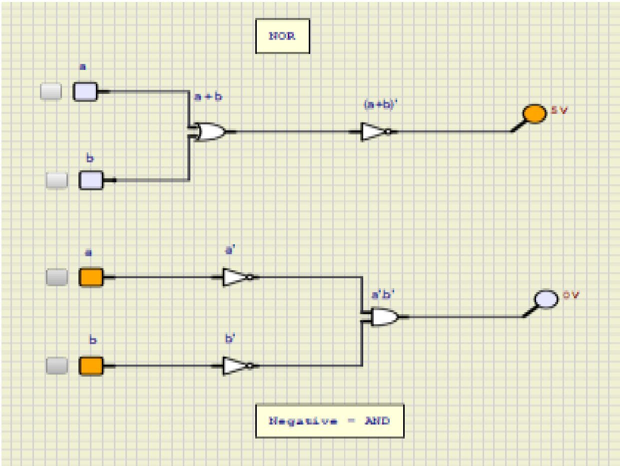
Verifying DeMorgan's Second Theorem using Truth Table

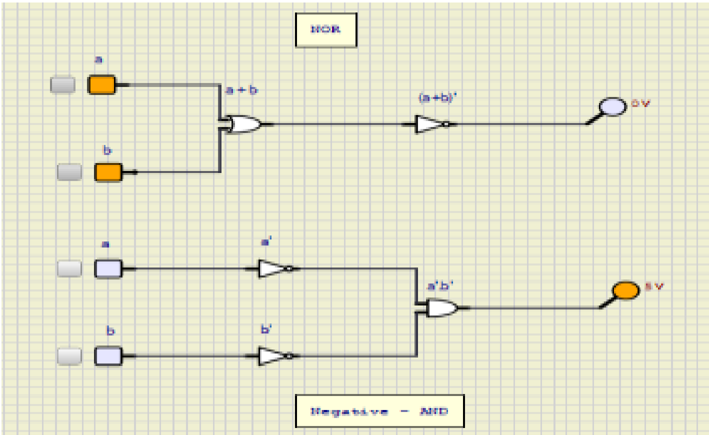
Inputs		Truth Table Outputs for Each Term				
B	A	A+B	A+B	A	B	A. B
0	0	0	1	1	1	1
0	1	1	0	0	1	0
1	0	1	0	1	0	0
1	1	1	0	0	0	0

Circuit Executions:









Results:

- Boolean expressions were successfully implemented in SimulIDE.
- Output matched expected values in the truth table.
- Circuit diagrams were verified and tested for various input combination.